

Convolutional neural networks, part 1

Weekend 3, Friday morning

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CAS Building ML/AI Applications, ETH Zurich

Friday 2 October 2026

Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

Segmentation and
detection



Generation



Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

$$f(\text{img}) = \begin{cases} \# \\ \text{Yes/No} \end{cases}$$

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

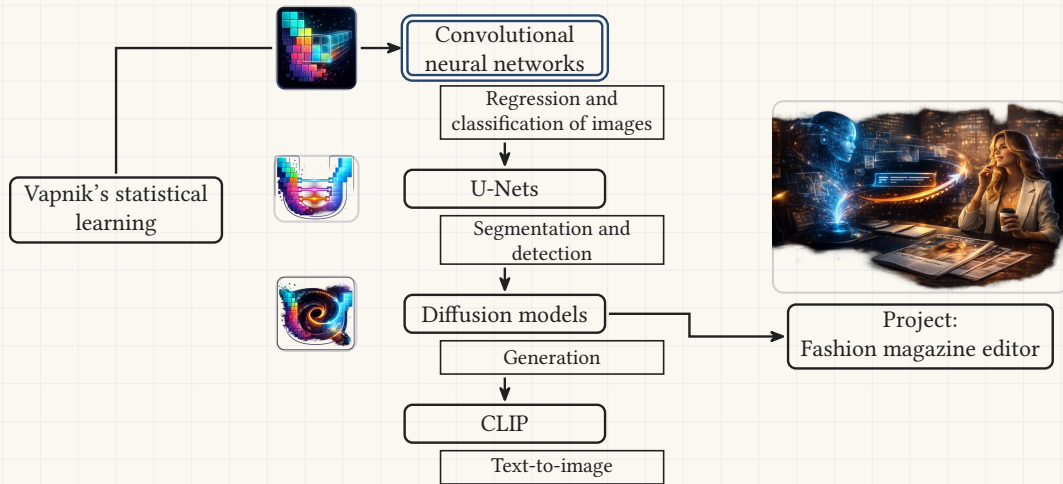
Segmentation and
detection



Generation



Agenda



1

Section 1

Statistical learning and neural networks

Statistical learning

How can I detect if a text was AI generated or not?

Statistical learning

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Collect
data

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

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$$\begin{array}{c} \uparrow \\ f(x) = y \begin{cases} 1 & \text{if } x \text{ is AI generated} \\ 0 & \text{otherwise} \end{cases} \\ \text{text} \end{array}$$

Statistical learning

How can I detect if a text was AI generated or not?

Collect data

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

Choose model

$$\hat{f}(x) = \text{NN}(x) = \hat{y}$$

$\hat{w}$, the parameters


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$$\text{for } i \leq n: \hat{y}_i = \hat{f}(x_i)$$

$$l_i = \mathcal{L}(y_i, \hat{y}_i)$$

$$l = \frac{1}{n} \sum_i l_i$$

$$f(x) = y \begin{cases} 1 & \text{if } x \text{ is AI generated} \\ 0 & \text{otherwise} \end{cases}$$

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$$\hat{w} = \hat{w} - \eta \nabla_{\hat{w}} l$$

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until



A curved arrow originates from the $\hat{f}(x)$ term in the model equation and points towards the $f(x)$ term in the prediction function equation.

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$\hat{w}$

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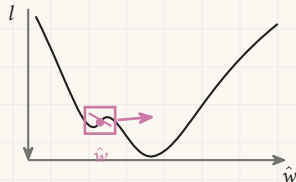
$$l = \frac{1}{n} \sum_i l_i$$

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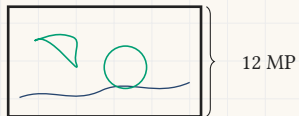
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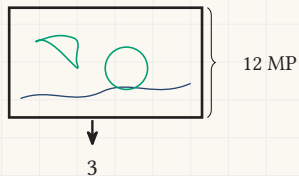
text



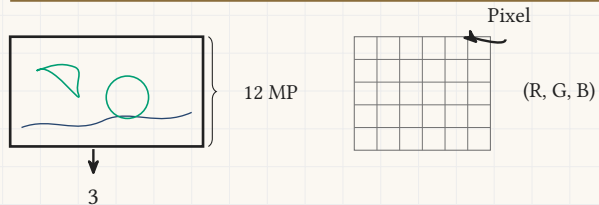
Images as pixel tensors



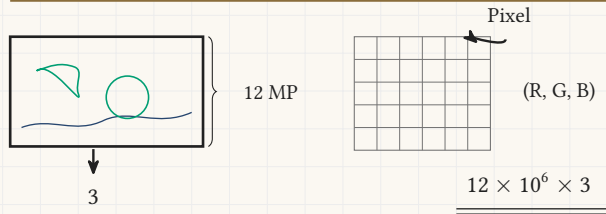
Images as pixel tensors



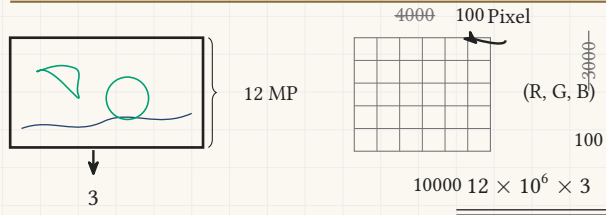
Images as pixel tensors



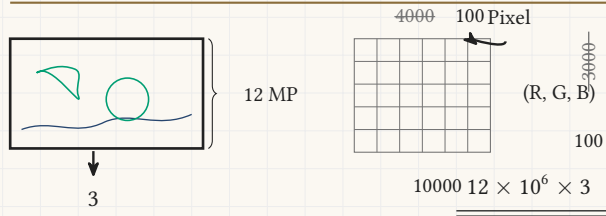
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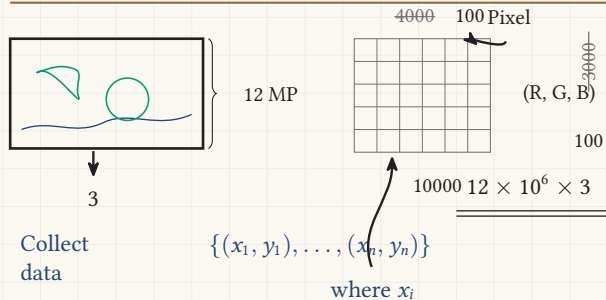
Images as pixel tensors



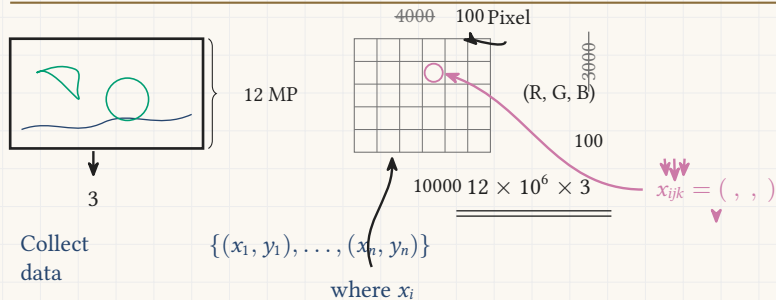
Collect
data

$\{(x_1, y_1), \dots, (x_n, y_n)\}$

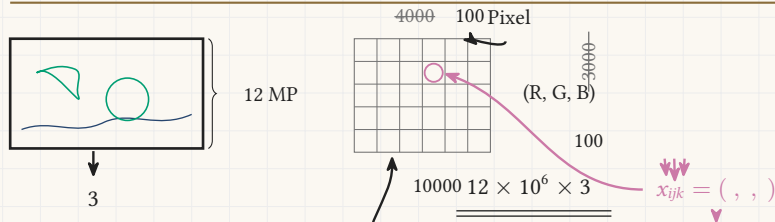
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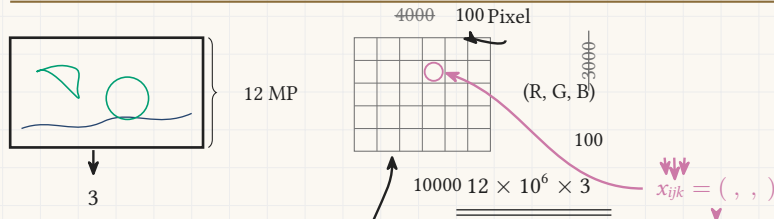
where x_i

Define
model

$$\hat{y}_i = \hat{f}(x_i)$$

$$= \hat{w}^T x_i + b$$

Images as pixel tensors



Collect
data

$$\{(x_1, y_1), \dots, (x_n, y_n)\}$$

where x_i

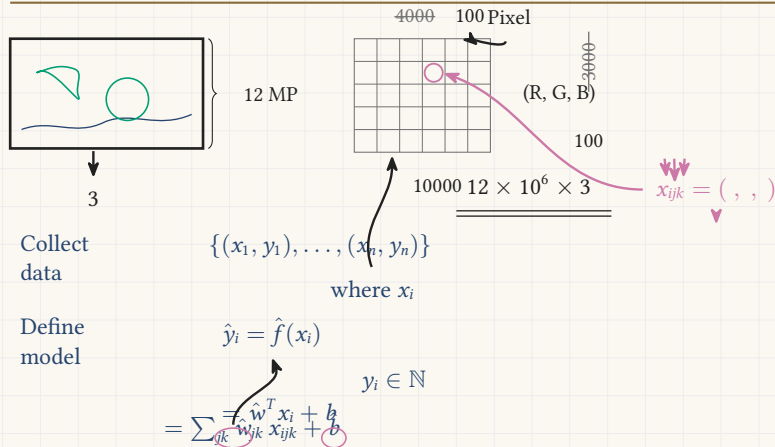
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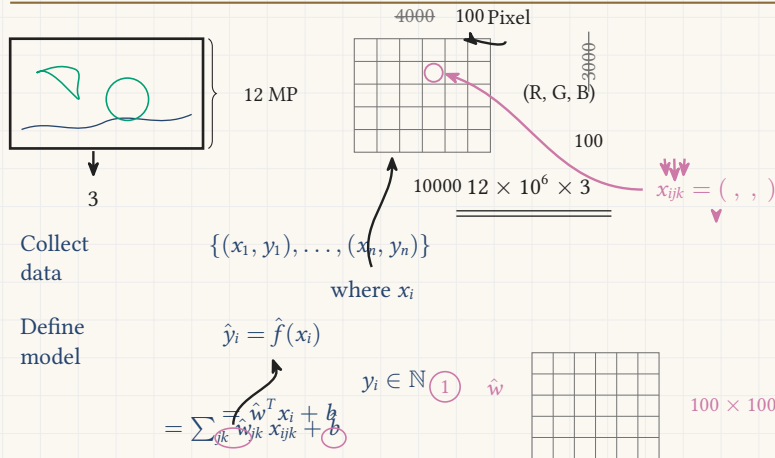
$$y_i \in \mathbb{N}$$

$$= \hat{w}^T x_i + b$$

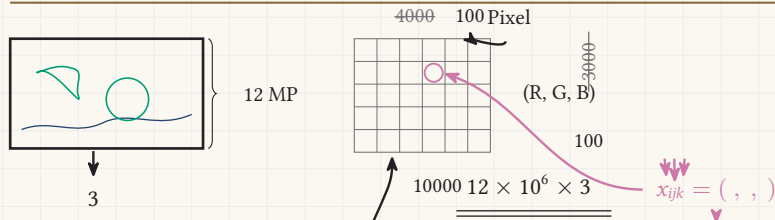
Images as pixel tensors



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Collect
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$$\{(x_1, y_1), \dots, (x_n, y_n)\}$$

where x_i

Define
model

$$\hat{y}_i = \hat{f}(x_i)$$

$$= \sum_{jk} \hat{w}_{jk} x_{ijk} + \hat{b}$$

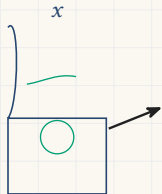
$$y_i \in \mathbb{N} \quad (1) \quad \hat{w}$$

(2)

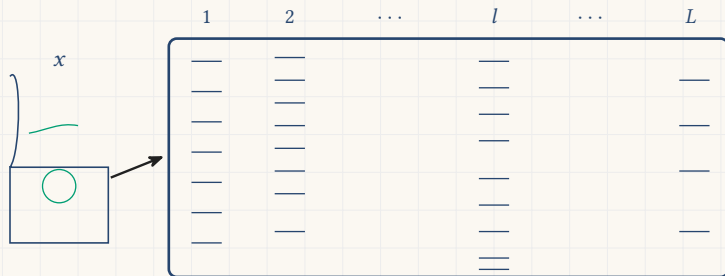


$$100 \times 100 \times 3 = 30000$$

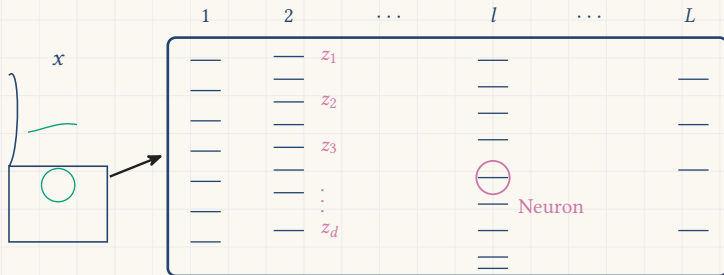
Neural networks



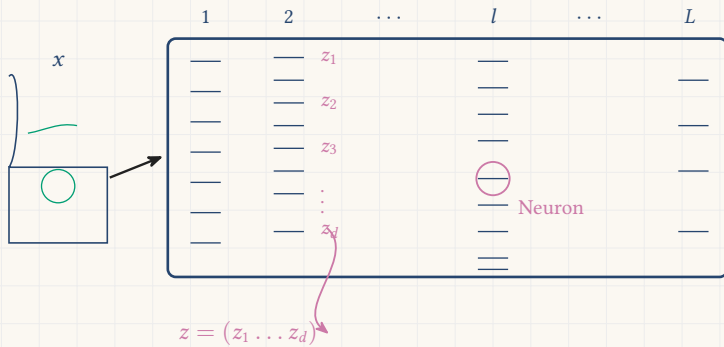
Neural networks



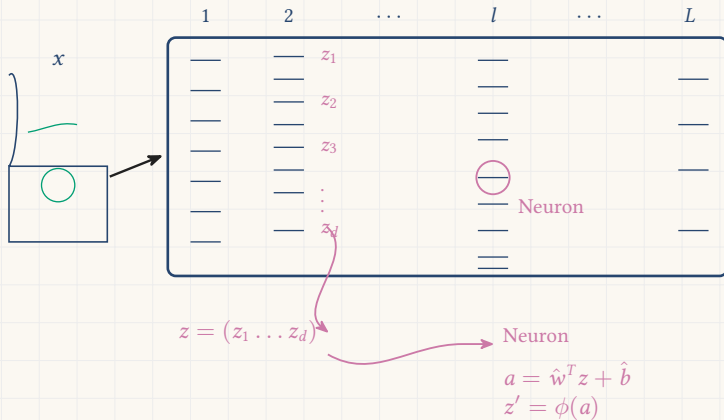
Neural networks



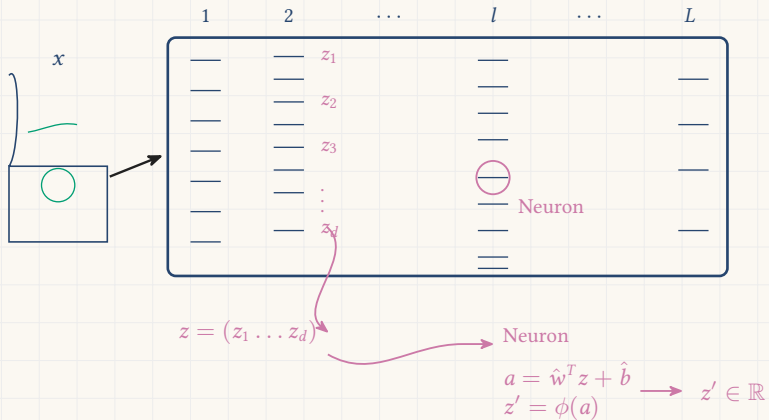
Neural networks



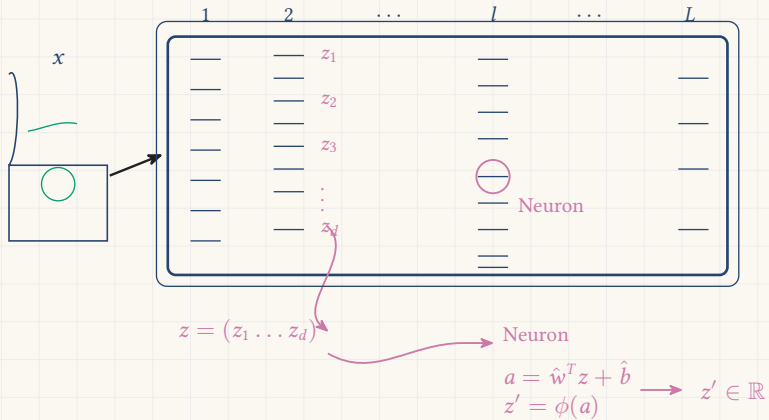
Neural networks



Neural networks



Neural networks

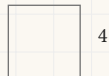
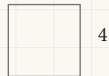
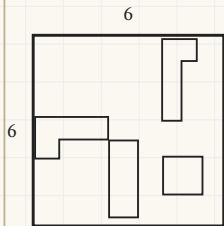


2

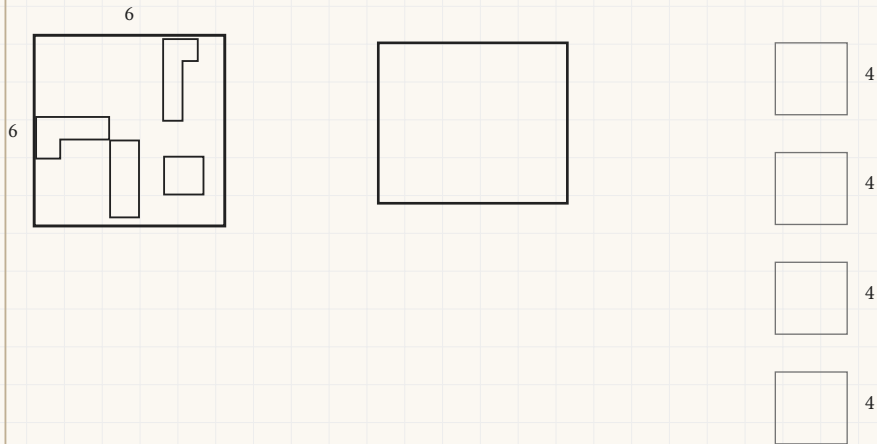
Section 2

Convolutional filters

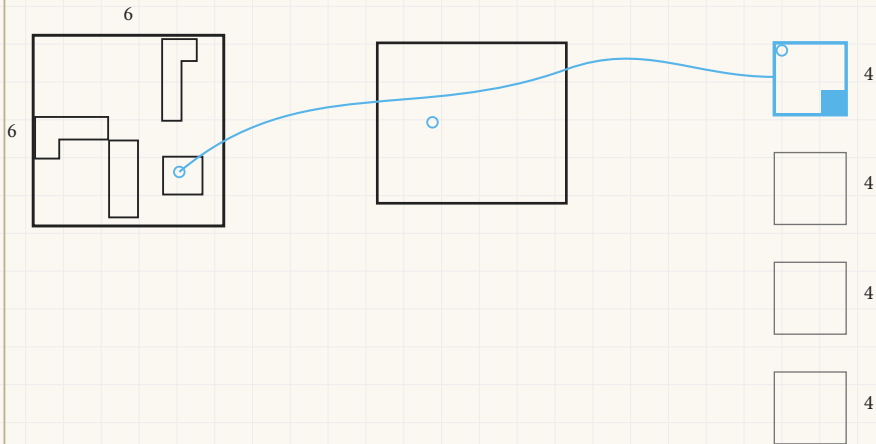
Filters find islands



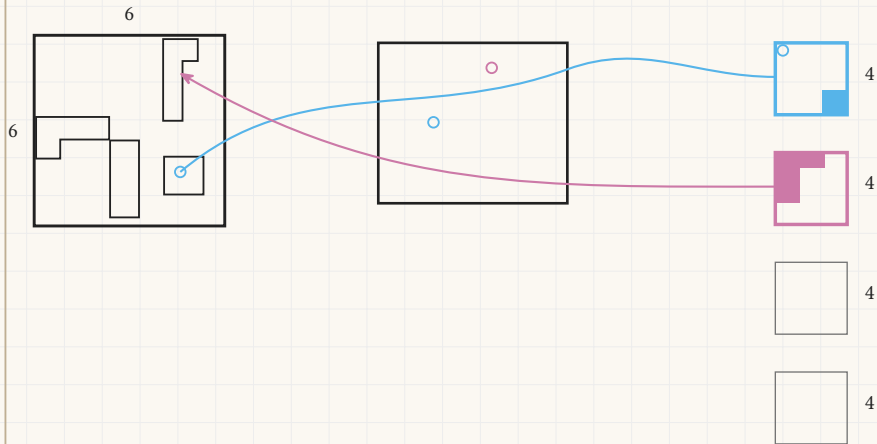
Filters find islands



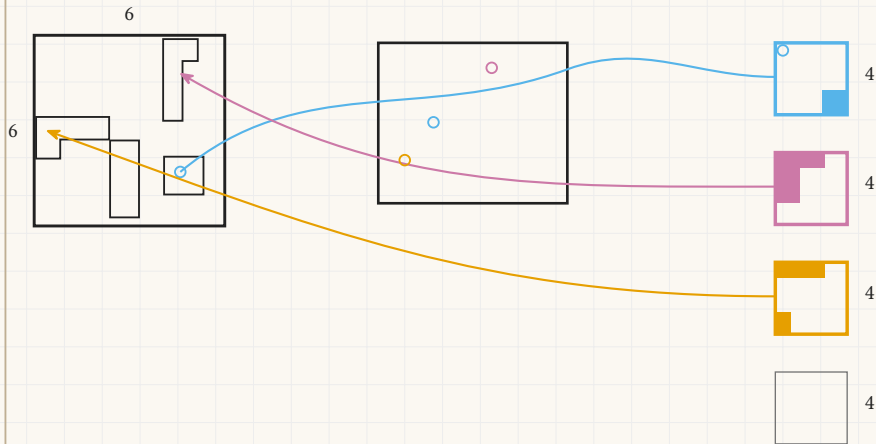
Filters find islands



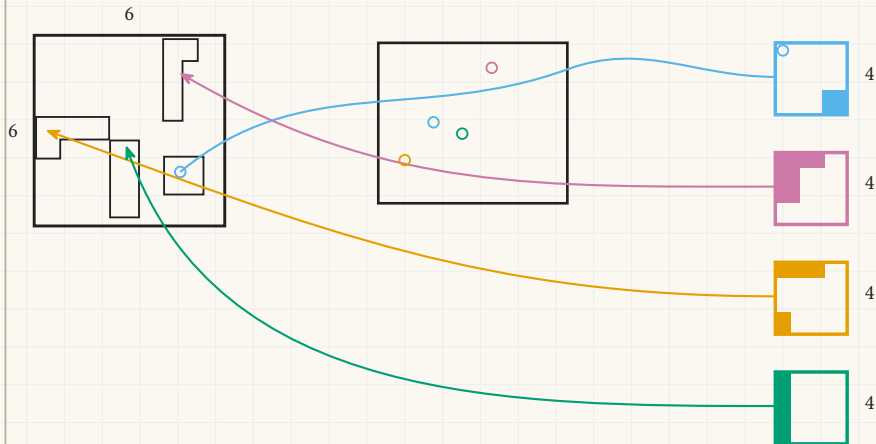
Filters find islands



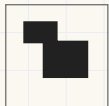
Filters find islands



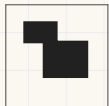
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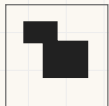
Recognising islands that look like a cross



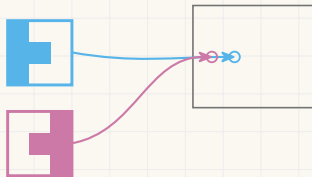
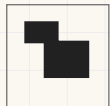
Recognising islands that look like a cross



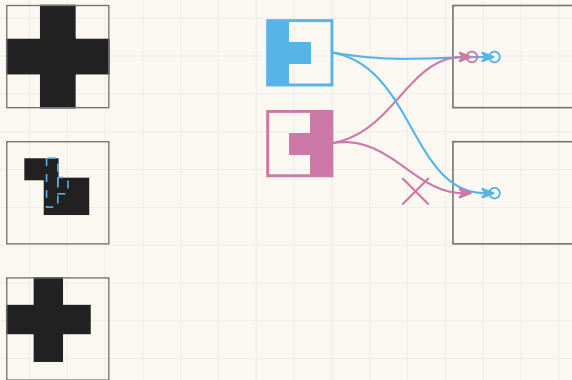
Recognising islands that look like a cross



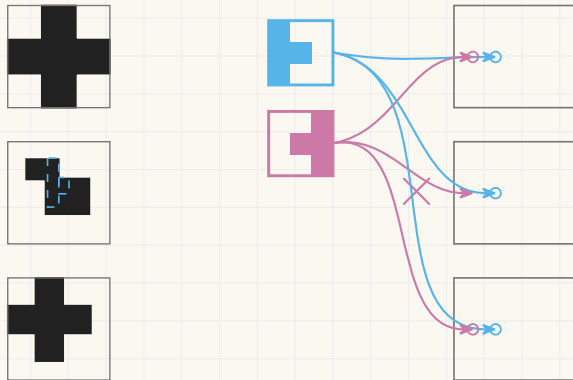
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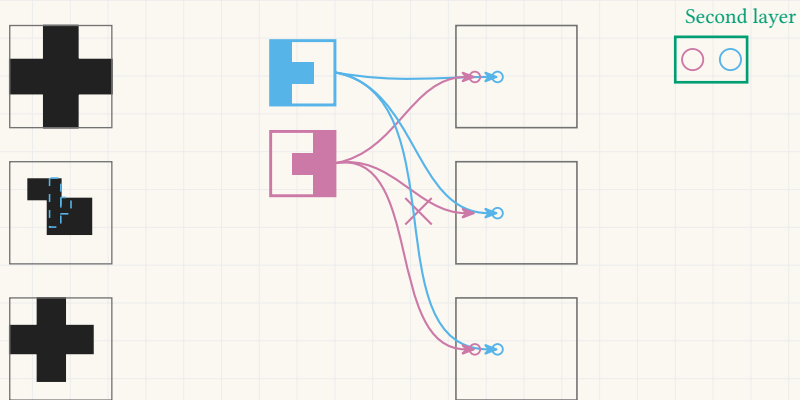
Recognising islands that look like a cross



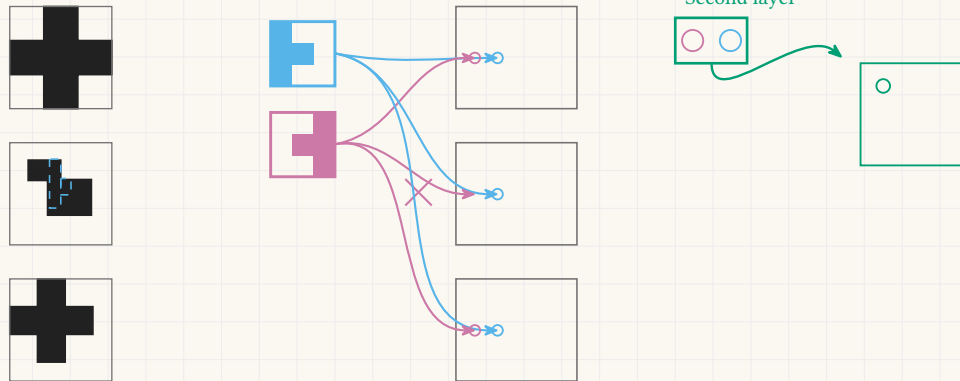
Recognising islands that look like a cross



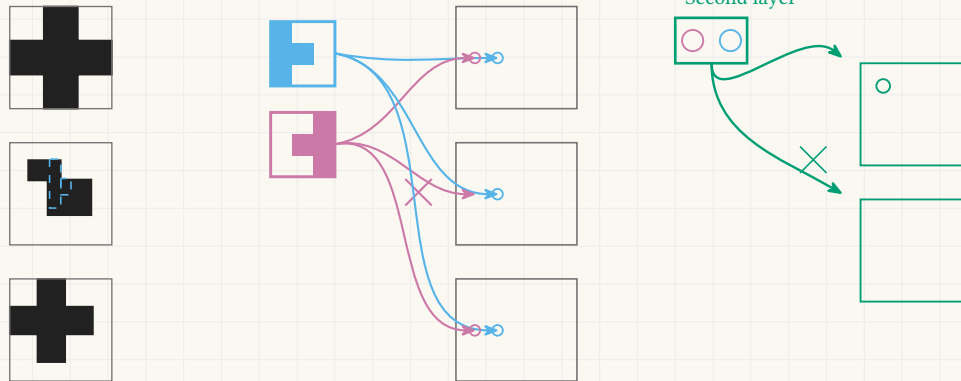
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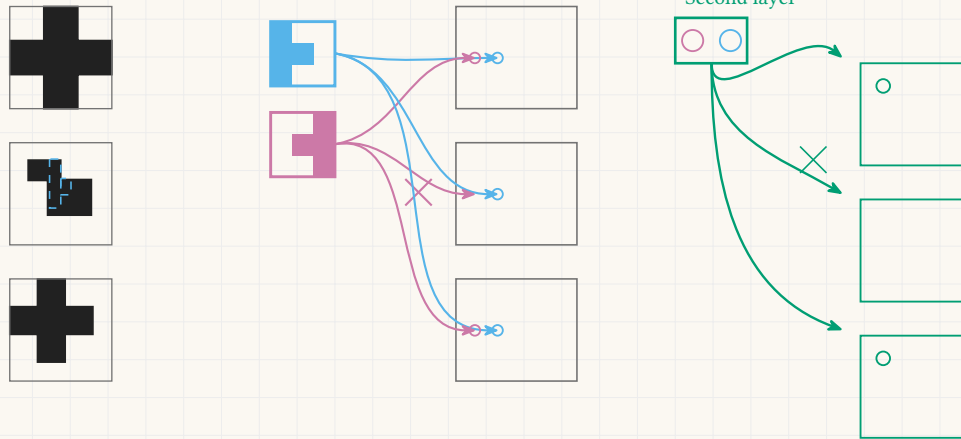
Recognising islands that look like a cross



Recognising islands that look like a cross



Recognising islands that look like a cross



How to compute activations



■ = 1
□ = 0



3x3

0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	1	1	1
0	0	0	0	1	0
0	0	0	0	0	0

How to compute activations



■ = 1
□ = 0



3x3

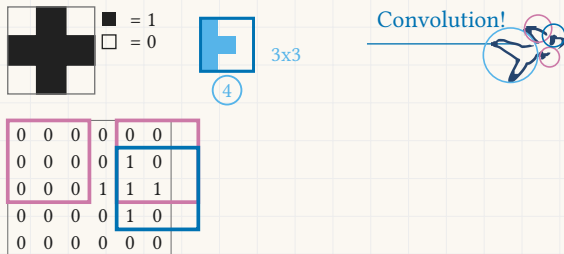
4

Convolution!



0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	1	1	1
0	0	0	0	1	0
0	0	0	0	0	0

How to compute activations



How to compute activations



■ = 1
□ = 0



3x3

4

Convolution!



0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	1	1	1
0	0	0	0	0	1	0
0	0	0	0	0	0	0

Desired output

0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

How to compute activations



■ = 1
□ = 0



3x3

4

Convolution!



Desired output

0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	1	1	1
0	0	0	0	1	0
0	0	0	0	0	0

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} * \begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} = \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \text{ Sum} \rightarrow 0$$

$$\begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{array} * \begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} = \begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{array} \text{ Sum} \rightarrow 2$$

How to compute activations



■ = 1
□ = 0



3x3

4

Convolution!



Desired output

0	0	0	0	0	0
0	0	0	0	1	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

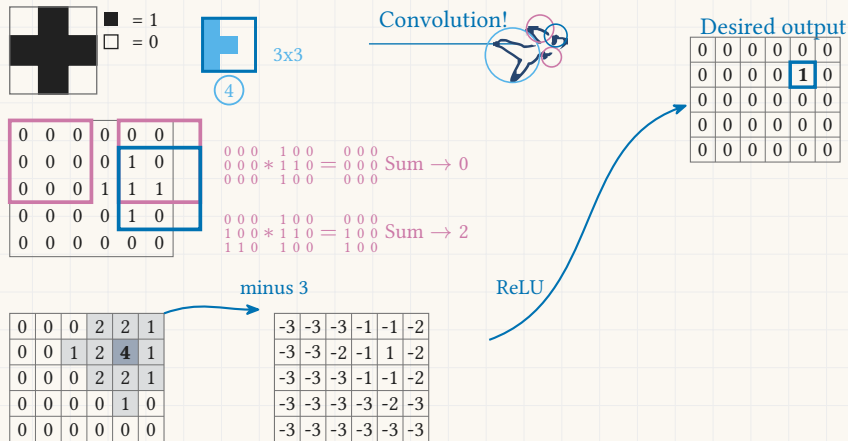
0	0	0	0	0	0	0
0	0	0	0	0	1	0
0	0	0	0	1	1	1
0	0	0	0	0	1	0
0	0	0	0	0	0	0

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} * \begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} = \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \text{ Sum} \rightarrow 0$$

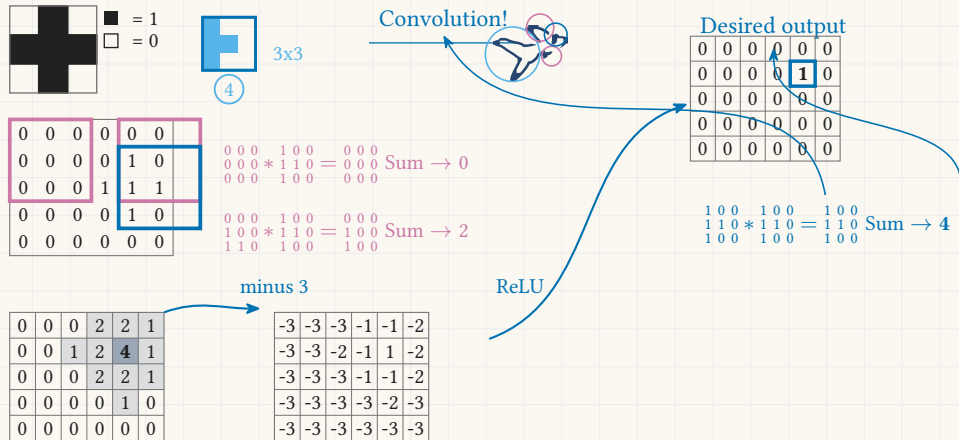
$$\begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{array} * \begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} = \begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{array} \text{ Sum} \rightarrow 2$$

0	0	0	2	2	1
0	0	1	2	4	1
0	0	0	2	2	1
0	0	0	0	1	0
0	0	0	0	0	0

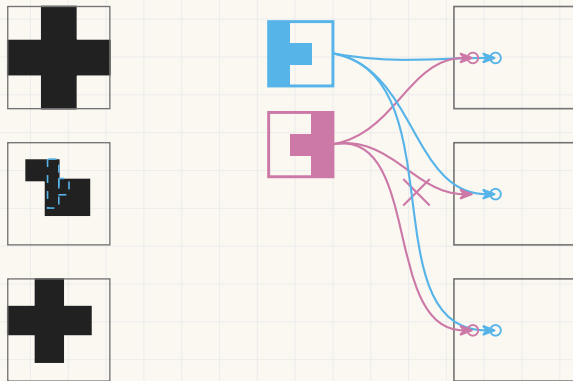
How to compute activations



How to compute activations

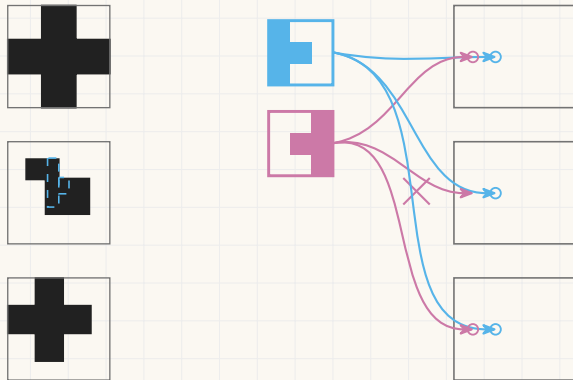


Recognising islands that look like a cross, with pooling

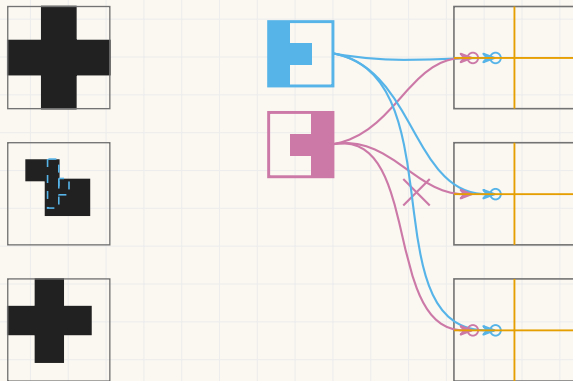


Recognising islands that look like a cross, with pooling

Pooling

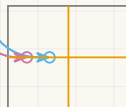
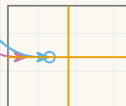
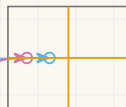


Recognising islands that look like a cross, with pooling



Pooling

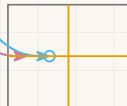
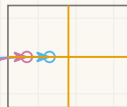
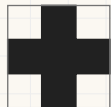
Recognising islands that look like a cross, with pooling



Pooling

1	0	1	0
0	0	0	0

Recognising islands that look like a cross, with pooling

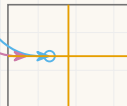
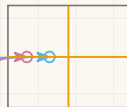
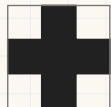


Pooling

1	0	1	0
0	0	0	0

1	0	0	0
0	0	0	0

Recognising islands that look like a cross, with pooling



Pooling

1	0	1	0
0	0	0	0

1	0	0	0
0	0	0	0

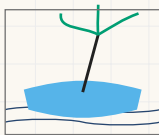
0	1	0	1
0	0	0	0

3

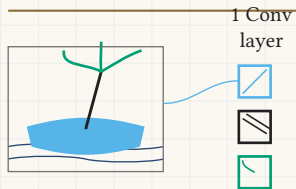
Section 3

The convolutional network

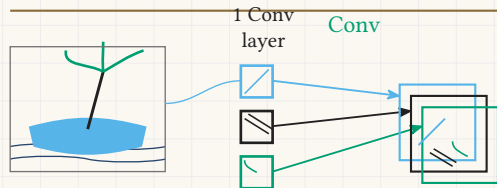
The convolutional network



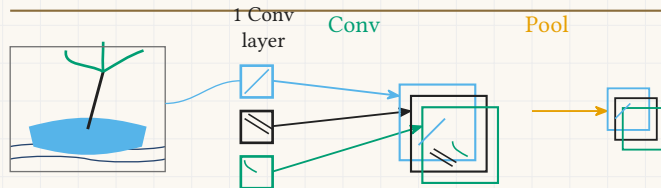
The convolutional network



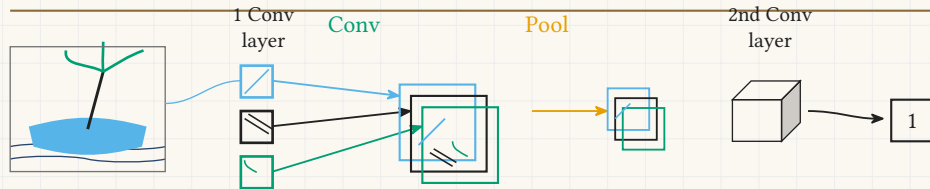
The convolutional network



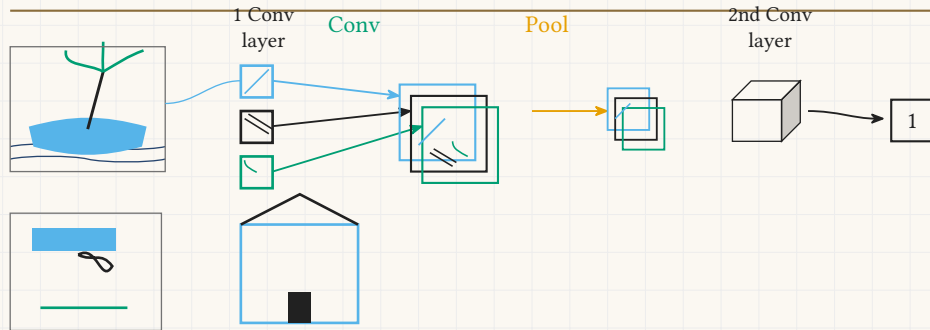
The convolutional network



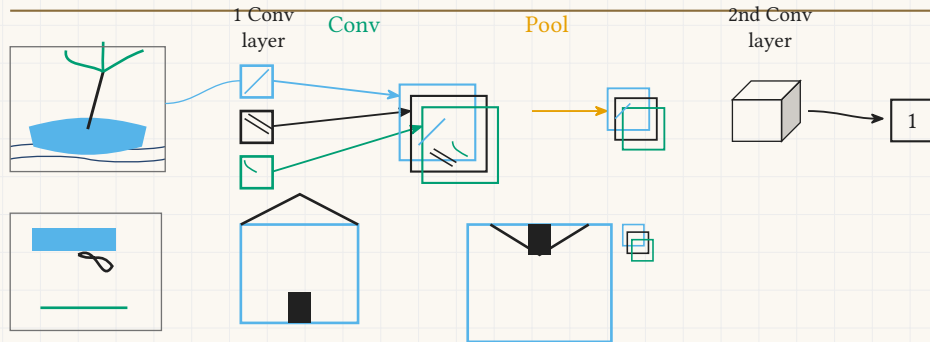
The convolutional network



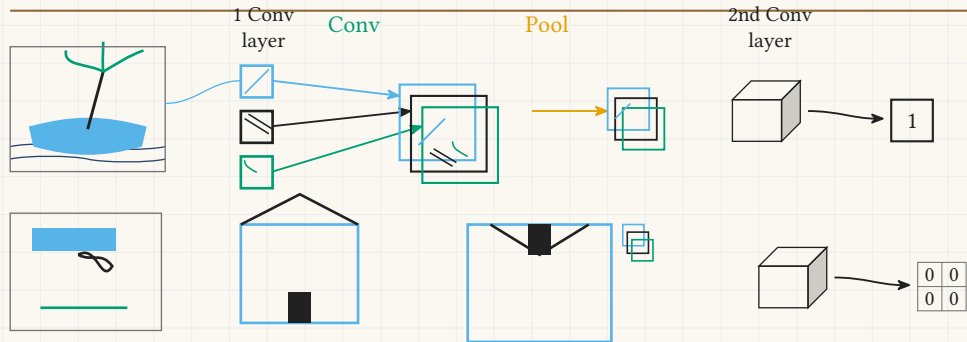
The convolutional network



The convolutional network



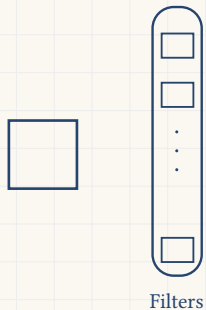
The convolutional network



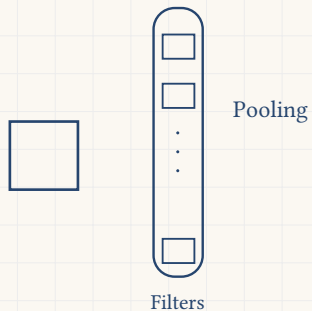
From filters to a decision



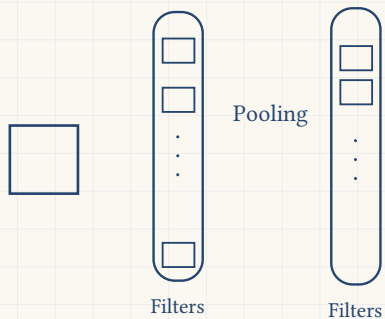
From filters to a decision



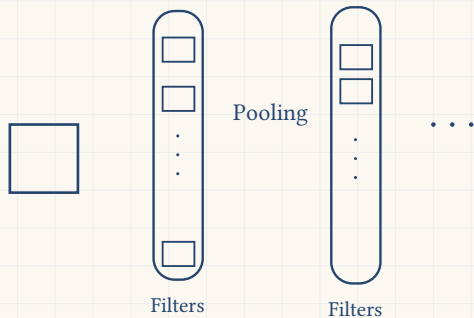
From filters to a decision



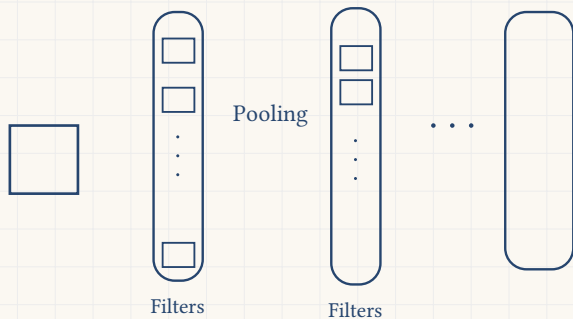
From filters to a decision



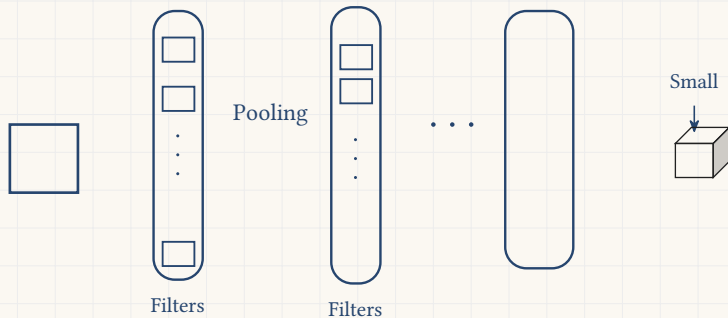
From filters to a decision



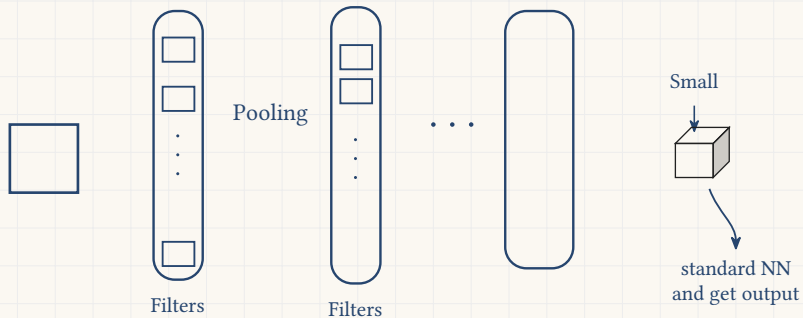
From filters to a decision



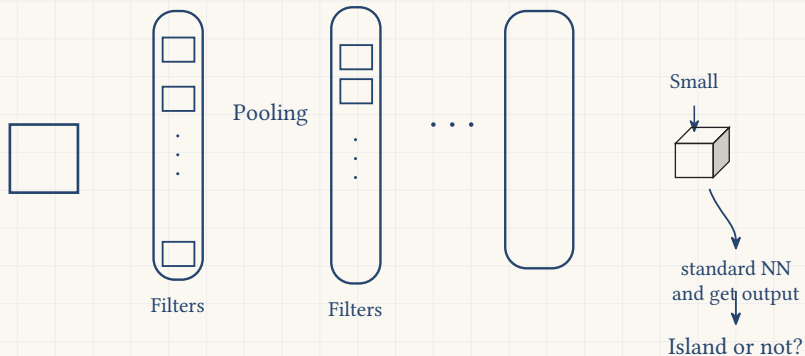
From filters to a decision



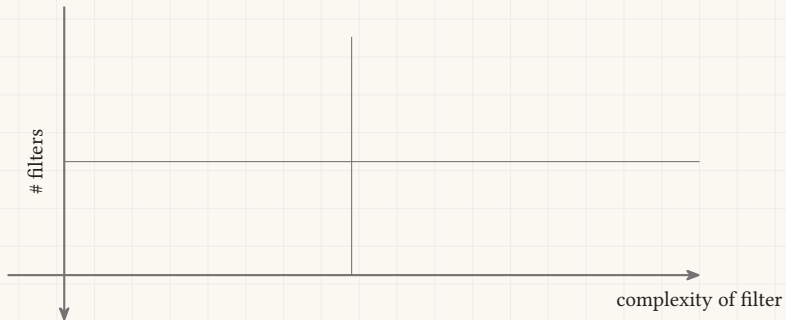
From filters to a decision



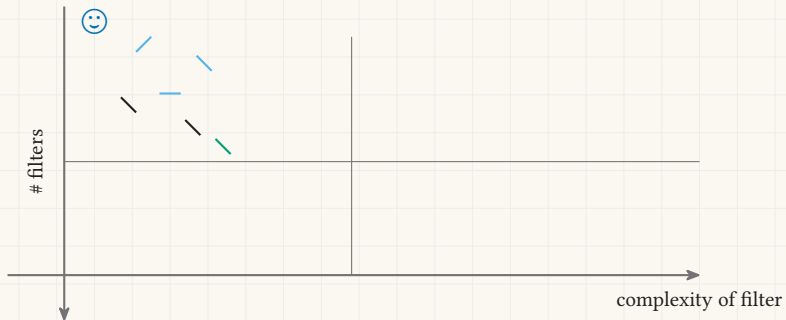
From filters to a decision



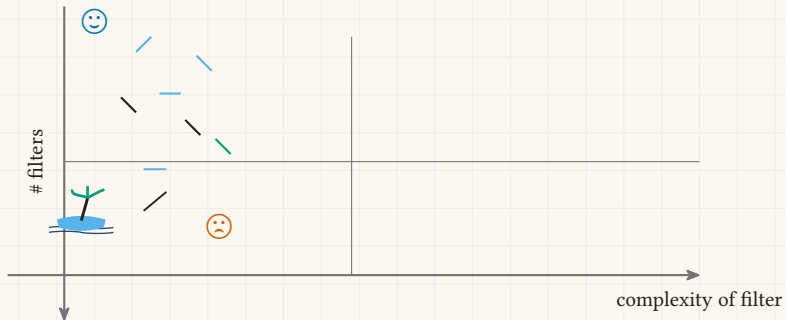
How many filters, how complex?



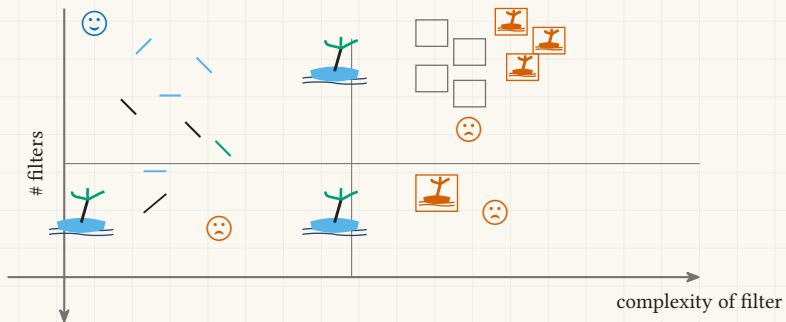
How many filters, how complex?



How many filters, how complex?



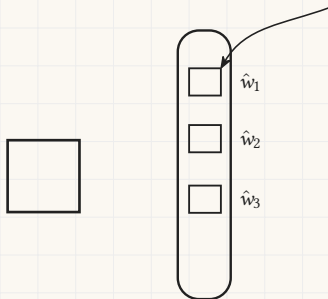
How many filters, how complex?



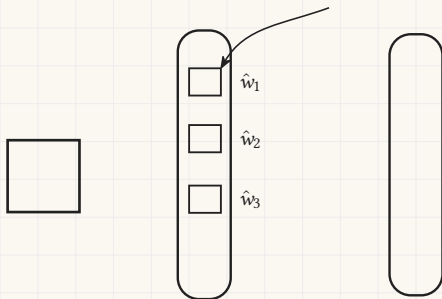
The filters are the parameters



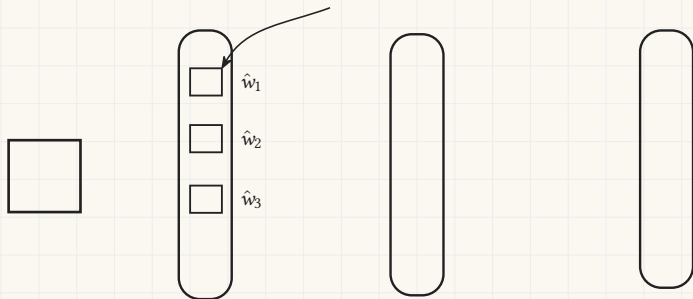
The filters are the parameters



The filters are the parameters



The filters are the parameters



A worked convolution

0	0	0	0	0	0
0	0	1	0	0	0
0	1	1	1	0	0
0	0	1	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

A worked convolution

0	0	0	0	0	0
0	0	1	0	0	0
0	1	1	1	0	0
0	0	1	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0



A worked convolution

✓

0	0	0	0	0	0
0	0	1	0	0	0
0	1	1	1	0	0
0	0	1	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0



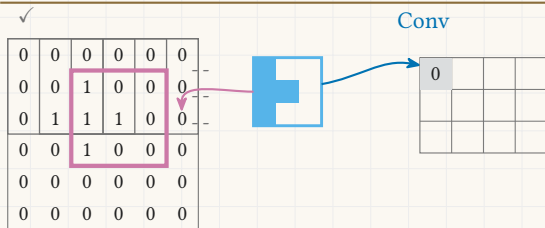
A worked convolution

✓

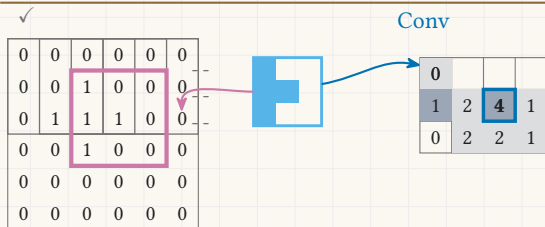
0	0	0	0	0	0	
0	0	1	0	0	0	
0	1	1	1	0	0	
0	0	1	0	0	0	
0	0	0	0	0	0	
0	0	0	0	0	0	



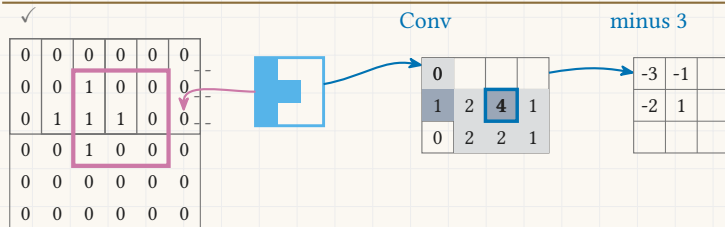
A worked convolution



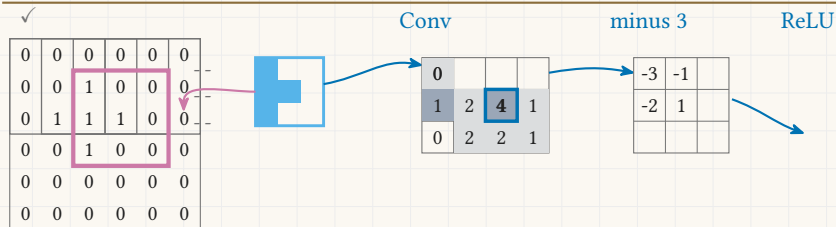
A worked convolution



A worked convolution



A worked convolution

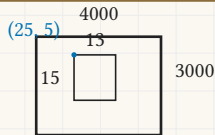


4

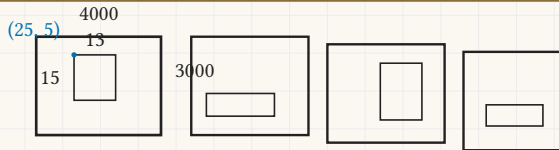
Section 4

How to discover the essential features

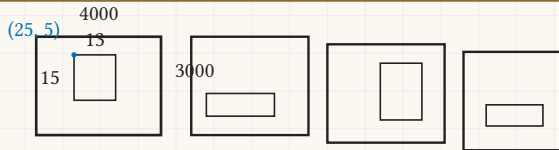
How to discover the essential features?



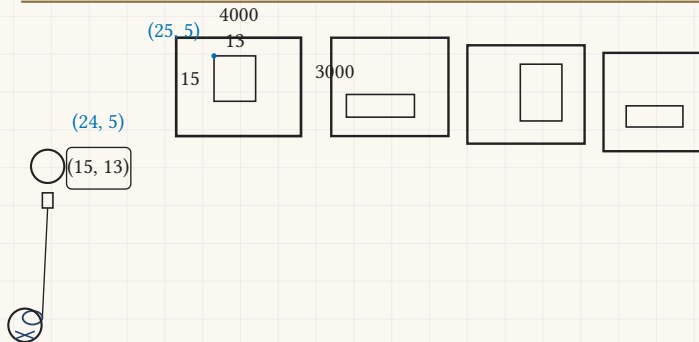
How to discover the essential features?



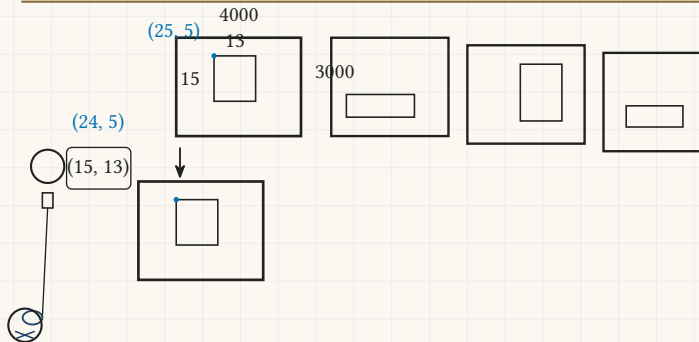
How to discover the essential features?



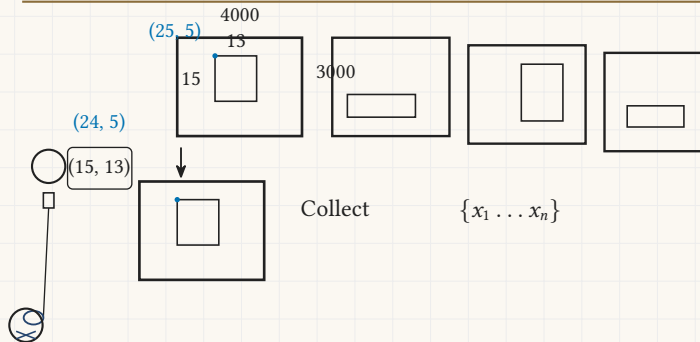
How to discover the essential features?



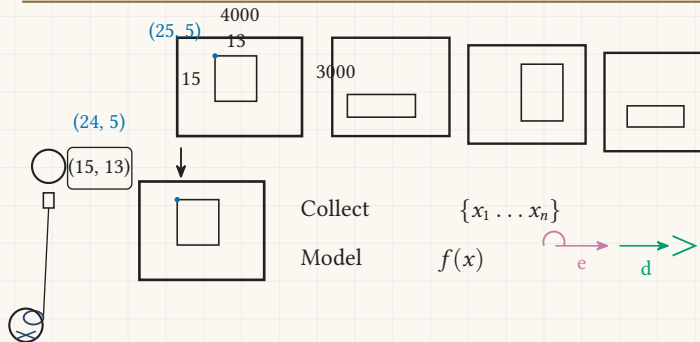
How to discover the essential features?



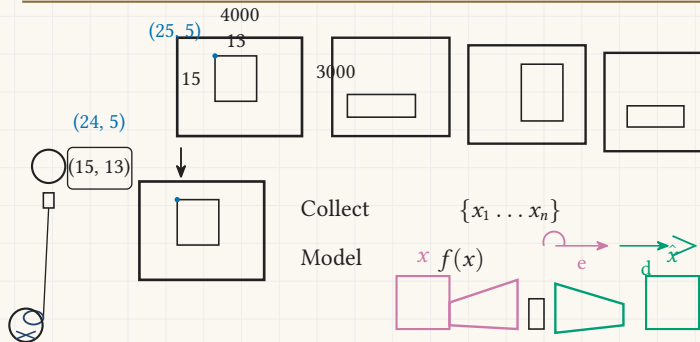
How to discover the essential features?



How to discover the essential features?



How to discover the essential features?



How to discover the essential features?

